



Faculty of Technology and Engineering

Computer Science and Engineering

Practical

Academic Year	:	2025-26	Semester	:	6
Course code	:	CSE312	Course name	:	Design of language processing

Practical - 5

1. Objective:

Implementation of a Lexical Analyzer for C Language Compiler.

2. Program Code:

Test 1:

```
int main() {
int a = 5 , 7H;
// assign value
char b = 'x';
/* return
value */
return a + b;
}
```

Test 2:

```
/* salary calculation*/
void main( )
{
long int bs , da , hra , gs;
//take basic salary as input
scanf("%ld",&bs);
//calculate allowances
da=bs*.40;
hra=bs*.20;
gs=bs+da+hra;
// display salary slip
printf("\nbs : %ld",bs);
printf("\nda : %ld",da);
printf("\nhra : %ld",hra);
printf("\ngs : %ld",gs);
}
```

```
}  
//function prototype  
void add ( int , int );  
void main()  
{  
int a , b;  
a = 10;  
b = 20;  
// function call  
add ( a , b );  
}  
void add ( int x , int y )  
{  
return x + y;  
}
```

Test 3:

```
// user defined data type  
struct student  
{  
int id;  
float cgpa;  
}  
void main()  
{  
student s;  
s.id = 10;  
s.cgpa = 8.7;  
}
```

3.Output:

```
debdootmanna@Mac.mshome.net ~ > College > Sem 6 by college > DLP > 5 main
> python3 lexical_analyzer.py test1.c
TOKENS
Keyword: int
Identifier: main
Punctuation: (
Punctuation: )
Punctuation: {
Keyword: int
Identifier: a
Operator: =
Constant: 5
Punctuation: ,
Constant: 7
Identifier: H
Punctuation: ;
Keyword: char
Identifier: b
Operator: =
String: 'x'
Punctuation: ;
Keyword: return
Identifier: a
Operator: +
Identifier: b
Punctuation: ;
Punctuation: }
```

```

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> python3 lexical_analyzer.py test2.c
TOKENS
Keyword: void
Identifier: main
Punctuation: (
Punctuation: )
Punctuation: {
Keyword: long
Keyword: int
Identifier: bs
Punctuation: ,
Identifier: da
Punctuation: ,
Identifier: hra
Punctuation: ,
Identifier: gs
Punctuation: ;
Identifier: scanf
Punctuation: (
String: "%ld"
Punctuation: ,
Operator: &
Identifier: bs
Punctuation: )
Punctuation: ;
Identifier: da
Operator: =
Identifier: bs
Operator: *
Operator: .
Constant: 40
Punctuation: ;
Identifier: hra
Operator: =
Identifier: bs
Operator: *
Operator: .
Constant: 20
Punctuation: ;
Identifier: gs
Operator: =
Identifier: bs
Operator: +
Identifier: da
Operator: +
Identifier: hra
Punctuation: ;
Identifier: printf
Punctuation: (
String: "\nbs : %ld"
Punctuation: ,
Identifier: bs
Punctuation: )
Punctuation: ;
Identifier: printf
Punctuation: (
String: "\nda : %ld"
Punctuation: ,
Identifier: da
Punctuation: )
Punctuation: ;
Identifier: printf
Punctuation: (

```

```
String: "\nhra : %ld"
Punctuation: ,
Identifier: hra
Punctuation: )
Punctuation: ;
Identifier: printf
Punctuation: (
String: "\ngs : %ld"
Punctuation: ,
Identifier: gs
Punctuation: )
Punctuation: ;
Punctuation: }
Keyword: void
Identifier: add
Punctuation: (
Keyword: int
Punctuation: ,
Keyword: int
Punctuation: )
Punctuation: ;
Keyword: void
Identifier: main
Punctuation: (
Punctuation: )
Punctuation: {
Keyword: int
Identifier: a
Punctuation: ,
Identifier: b
Punctuation: ;
Identifier: a
Operator: =
Constant: 10
Punctuation: ;
Identifier: b
Operator: =
Constant: 20
Punctuation: ;
Identifier: add
Punctuation: (
Identifier: a
Punctuation: ,
Identifier: b
Punctuation: )
Punctuation: ;
Punctuation: }
Keyword: void
Identifier: add
Punctuation: (
Keyword: int
Identifier: x
Punctuation: ,
Keyword: int
Identifier: y
Punctuation: )
Punctuation: {
Keyword: return
Identifier: x
Operator: +
Identifier: y
Punctuation: ;
Punctuation: }
```

```
debdoottmanna@Mac.mshome.net ~ > College > Sem 6 by college > DLP > 5 ↩main  
> python3 lexical_analyzer.py test3.c  
TOKENS  
Keyword: struct  
Identifier: student  
Punctuation: {  
Keyword: int  
Identifier: id  
Punctuation: ;  
Keyword: float  
Identifier: cgpa  
Punctuation: ;  
Punctuation: }  
Keyword: void  
Identifier: main  
Punctuation: (  
Punctuation: )  
Punctuation: {  
Identifier: student  
Identifier: s  
Punctuation: ;  
Identifier: s  
Operator: .  
Identifier: id  
Operator: =  
Constant: 10  
Punctuation: ;  
Identifier: s  
Operator: .  
Identifier: cgpa  
Operator: =  
Constant: 8.7  
Punctuation: ;  
Punctuation: }
```